

LECTURE L15-16 • MODULE 3 • CO3

Data quality dimensions

Week 8 | Slides: DATA209_Module_3_Data_Quality_Engineering.pptx, pages 4-11 | Laboratory: P15-16

TOPICS COVERED

- Accuracy, completeness, consistency, validity
- Integrating data from multiple sources
- Duplicate detection, inconsistent encoding, validation rules

Cleaning is inference

Every cleaning decision is a modelling decision made early, with less evidence and usually without a record.

Imputing a mean asserts something about the missing values. Deleting an outlier asserts it was an error. Neither is neutral, and both must be written down.

Six dimensions

DIMENSION	QUESTION
Accuracy	do the values reflect reality? A blood pressure of 0 is precise, recorded and wrong
Completeness	is anything absent that should be present — including <i>disguised</i> absence?
Consistency	do values agree across records and sources?
Validity	do values conform to type, range, allowed set and cross-field rules?
Uniqueness	is each real-world entity represented once?
Timeliness	is the data current enough, and is every record from the same period?

Audit in this order: Validity failures often explain completeness failures, which often explain accuracy failures.

Profiling

Per column: dtype, nulls, **zeros**, unique count, min, max, top values. Then exact duplicate rows, then duplicates on the business key alone. For numeric columns check the range against what is physically possible; for categorical columns list the full value set and hunt for near-duplicates; for dates check range, ordering and parse failures.

The zeros column is the one that finds disguised missingness — see L17-18.

Multi-source integration

PROBLEM	SYMPTOM	RESPONSE
key mismatch	far fewer rows after the join	normalise keys: trim, case-fold, check type
granularity mismatch	row count explodes	aggregate the finer side first
schema drift	same name, different meaning or unit	rename explicitly; convert units
overlapping records	the same entity in both sources	deduplicate after the join
conflicting values	sources disagree	define precedence in advance; record which won

Always compare row counts before and after a join. **An unexplained change in row count is a bug until proven otherwise.**

Duplicates and encodings

Duplicates come in three grades: **exact** (identical across all columns), **key** (same entity, differing fields — the harder and more common case), and **fuzzy** ("Ramesh Kumar" vs "Ramesh Kumar", needing string distance). Decide which record survives: most recent, most complete, or a merge.

Inconsistent encodings: case and whitespace; mixed units in one column; DD/MM/YYYY against MM/DD/YYYY, silently wrong for the first twelve days of a month; sentinels such as 999, -1, NA, N/A, '' and unknown all meaning missing. **Standardise before you count** — every frequency table computed on unstandardised categories is wrong.

Validation rules

Write them from domain knowledge *before* looking at the data, then measure the violations. Type rules (age is an integer), range rules (glucose between 40 and 400), set rules (VisitorType is one of three known values) and cross-field rules (discharge is not before admission; a duration cannot be positive when the page count is zero).

The **cleaning log** — finding, rows affected, action, justification — is a deliverable, not a courtesy. It carries most of the marks in Assignment 2.

KEY TERMS

Data profiling	Accuracy	Completeness	Consistency	Validity	Uniqueness	Timeliness
Business key	Fuzzy duplicate	Sentinel value	Cleaning log			

COMMON ERRORS TO AVOID

- Joining two sources without comparing row counts before and after
- Counting categories before standardising case and whitespace
- Deriving validation rules from the data instead of from the domain
- Cleaning without recording what was changed and why

READINGS — ALL FREE TO ACCESS

1. pandas — merge, join and concatenate
https://pandas.pydata.org/docs/user_guide/merging.html
2. Kaggle Learn — Data Cleaning
<https://www.kaggle.com/learn/data-cleaning>
3. Berkeley Data 100 — data cleaning notes
<https://ds100.org/course-notes/>

TEST YOURSELF

1. Your join turns 10,000 rows into 41,000. What is the most likely cause?
2. Give a cross-field validity rule for an e-commerce session table.
3. Why must standardisation precede a frequency table?
4. Which quality dimension does a stale exchange-rate column violate?